CHAPTER 1
REVIEW

KEY QUESTIONS AND TERMS

1. b **DOK 1**
2. d **DOK 1**
3. c **DOK 1**
4. d **DOK 1**
5. A *population* is a group of individuals that belong to the same species and live in the same area; a *community* is a group composed of different populations that live together in a defined area; an *ecosystem* is a community together with its physical environment. **DOK 1**
6. observation and measurement, experimentation, and modeling **DOK 1**
7. Global systems and phenomena are difficult (and often impossible) to examine through real-life experiments. Models help scientists research and explain these processes without affecting the natural world. **DOK 2**
8. Sample answer: The three parts of the Understanding Global Change model are causes of global change, processes in Earth's global systems, and how processes respond to causes of global change. Three processes that are related to each other are biotic factors, abiotic factors, and populations. For example, consider a population of antelope. Biotic factors such as predators (lions) or abiotic factors such as natural disasters (droughts) can reduce the size of populations. Populations of antelope can also affect biotic factors by overgrazing land and ruining plant life. **DOK 2**
9. c **DOK 1**
10. b **DOK 1**
11. c **DOK 1**
12. Climate is the patterns and averages of temperature, precipitation, clouds, and wind over many years for a location, whereas weather is the short-term changes in precipitation, temperature, clouds, and wind from day to day. **DOK 2**
13. The main factor that accounts for the unequal distribution of heat is the angle of incoming solar radiation. Equatorial regions receive more solar heat and are hotter, while polar regions receive much less solar heat and are colder. Other factors that affect the distribution include mountain ranges, oceans, and ocean currents. **DOK 2**
14. Wind is caused by uneven heating of the Earth's atmosphere. Air that is heated rises, and cool air sinks. This shifting causes air to flow in global wind patterns. **DOK 1**
15. The path of an ocean current is affected by global surface winds, the rotation of Earth, the presence of continents, and water temperature. **DOK 1**
16. Wind encountering a mountain range rises up the side of the mountain, cools, and loses any moisture as precipitation. On the other side of the range, the air descends, warms, and dries out. **DOK 2**

CHAPTER 1
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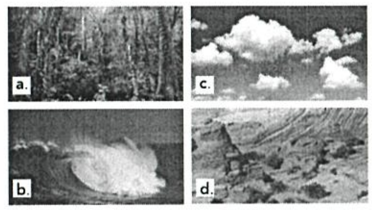
KEY QUESTIONS AND TERMS

1.1 Introduction to Global Systems

EP&CIIb, EP&CIIIa

1. The study of the complex system of interactions that sustain life on the planet is
a. zoology. c. chemistry.
b. ecology. d. economics.
2. Which photo represents the geosphere?



3. Nonliving factors of an environment are
a. biotic. c. abiotic.
b. bacteria. d. plankton.
4. Which of the following includes parts of the other three?
a. atmosphere. c. hydrosphere.
b. geosphere. d. biosphere.
5. Compare the terms *population*, *community*, and *ecosystem*.
6. What are three general techniques used to study ecology?
7. How can models be useful in explaining global system processes and phenomena?
8. Name and describe the three main parts of the Understanding Global Change model. Select three processes or phenomena explored in this chapter and describe how they are related to each other.

1.2 Climate, Weather, and Life

HS-LS2-2, HS-ESS2-4, EP&CIIIa

9. The climate zone closest to the equator is
a. polar. c. tropical.
b. temperate. d. torrid.
10. Average temperatures, precipitation, wind patterns, and extreme events in an area define its
a. geosphere. c. weather.
b. climate. d. atmosphere.
11. Concentrations of several heat-trapping gases in the atmosphere produce
a. radiation.
b. solar energy.
c. the greenhouse effect.
d. the hydrosphere.
12. How is climate different from weather?
13. What accounts for the unequal distribution of heat between the equator and the poles?
14. What phenomena shape and drive global wind patterns?
15. What factors affect the path of an ocean current?
16. How do mountain ranges affect climate?
17. What are some of the long-term non-human causes of global change?

1.3 Biomes and Aquatic Ecosystems

HS-LS2-2

18. The biome that supports more species than all other biomes is the
a. savannah.
b. temperate grassland.
c. boreal forest.
d. tropical rain forest.
19. Taiga is a synonym for the
a. boreal forest.
b. temperate woodland.
c. tropical dry forest.
d. desert.
20. Which variable do scientists use to divide the open ocean into two zones?
a. salinity
b. latitude
c. light
d. oxygen
21. Which biome is characterized by permafrost?
22. What factors describe aquatic ecosystems?
23. What is the ocean zone in which photosynthesis cannot occur?
24. All estuaries are wetlands, but not all wetlands are estuaries. Explain.
25. Where does the most photosynthesis on Earth occur?

1 Chapter 1 The Biosphere

17. Long-term natural causes of climate change include changes in solar radiation output, changes in Earth's orbit, volcanic activity, and meteorite impacts. **DOK 1**
18. d **DOK 1**
19. a **DOK 1**
20. c **DOK 1**
21. tundra **DOK 1**
22. salinity, depth, temperature, rate of flow, dissolved nutrients **DOK 1**
23. aphotic zone **DOK 1**
24. A wetland is an area that is covered with fresh water or has fresh water very near the surface during some part of the year. An estuary is an aquatic environment where fresh water meets salt water. **DOK 2**
25. In the photic zone of the open ocean **DOK 1**

CRITICAL THINKING

HS-LS2-2, HS-ESS2-4, EP&CIIb, EP&CIIIa

26. Evaluate Claims One of your classmates claims that most photosynthesis occurs in biomes with lots of trees. Are they correct? If so, justify their answer with reasoning and evidence. If not, suggest how to alter their claim and provide them with new reasoning and evidence.

27. SEP Plan an Investigation Ecologists have discovered that the seeds of many plants that grow in forests cannot germinate unless they have been exposed to fire. Design an experiment to test whether a particular plant has seeds with this requirement. Include your hypothesis statement, a description of both the control and experimental groups, and an outline of your procedure.

28. Construct an Argument One friend says biotic factors are more important than abiotic factors. Another friend says abiotic factors are more important. What do you think? Defend your position using examples from a specific biome of your choice.

29. Construct an Explanation A plant grower has a greenhouse where she grows plants in the winter. The greenhouse is exposed to direct sunlight, and by early spring, often gets too hot for the plants. She paints the outside of the glass with a chalky white paint, and the temperature remains at more acceptable levels. Explain why this solution works.

30. Integrate Information Although the amount of precipitation is low, most parts of the tundra are very wet during the summer. How would you explain this apparent contradiction?

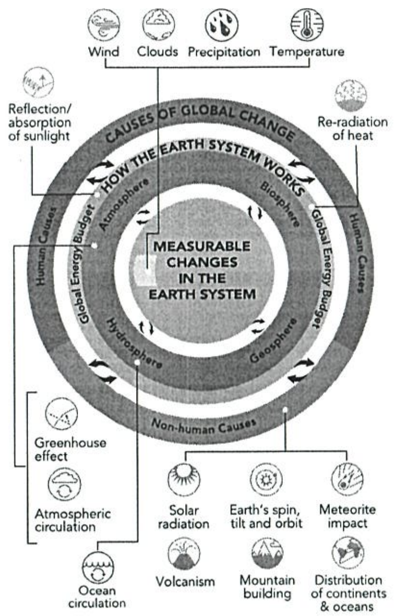
31. CCC Identify Patterns Coyotes can live in both temperate grasslands and temperate woodland biomes. Describe two adaptations that might enable coyotes to tolerate conditions in both biomes.

32. Construct an Explanation The greenhouse effect is an important part of Earth's climate. How could you create a model of the greenhouse effect? Consider the different elements that are involved in regulating the greenhouse effect, as well as the role that humans have had on the process.

33. Form a Hypothesis The deep ocean is in the aphotic zone and is very cold. Suggest some unique characteristics that enable animals to live there.

34. Communicate Information A developer has proposed filling in a salt marsh to create a coastal resort. What positive and negative effects do you think this proposal would have on wildlife and local residents? Would you support the proposal?

Use the Understanding Global Change model to answer questions 35–37.



35. CCC Use Models How are processes in the How the Earth System Works category related to Measurable Changes in the Earth System? Describe at least one relationship between two topics shown in the model.

36. Synthesize Information Consider the processes Atmospheric Circulation and Ocean Circulation in the "How the Earth System Works" ring. Choose three other processes or phenomena in the model and explain how each is related to Atmospheric circulation and Ocean circulation.

37. SEP Evaluate Models What does the model communicate about how processes and phenomena shape global systems? What have you learned about these topics that is not explained in the model?

CRITICAL THINKING

26. Sample answer: The classmate is incorrect. Most of the photosynthesis on our planet occurs in the oceans, which cover a much bigger percentage of the planet than forests. Small phytoplankton in the photic zones carry out photosynthesis. **DOK 2**

27. Sample answer: **Hypothesis:** Seeds from Plant X require fire to germinate.

Experimental group: seeds resting on a tray of soil covered with straw that is burned and then watered regularly after

Control: seeds resting on a tray of soil covered with straw that is watered regularly at the same time as is the experimental tray

Procedure: Use natural soil in five control and five experimental trays; add seeds and equal amounts of straw. Burn the straw in the trays in a fume hood. Water the soil in both sets of trays at 1-week intervals, and provide the same temperatures and light cycles to both. Record the number of germinating seeds in each tray. **DOK 4**

28. Both biotic and abiotic factors are important to the study of ecology and understanding ecosystems. For example, in a lake or stream, the quality of the water (an abiotic factor) affects the numbers and variety of organisms that live there. The organisms (biotic factors) also affect one another, such as by feeding relationships. **DOK 3**

29. The white paint reflects sunlight, which reduces the amount of solar energy that the greenhouse receives. **DOK 2**

30. The tundra is covered in snow and ice during the winter. When temperatures become warmer, the snow and ice melt, and the tundra remains damp through the summer. **DOK 2**

31. Sample answers: ability to tolerate seasonal temperature and precipitation changes, ability to travel on uneven ground. **DOK 3**

32. Sample answer: The greenhouse effect could be modeled by enclosing air in a tight space. This can be done by placing plastic wrap over a plastic container. Then, by shining a light on the plastic container, the air inside the container should heat up because the plastic wrap traps the heat, much like greenhouse gases trap heat in Earth's atmosphere. **DOK 2**

33. Sample answer: They may possess somewhat similar adaptations to land animals that live in frigid temperatures: round bodies that have smaller surface area to volume ratios, fat layers, and slow movements. **DOK 3**

34. The coastal salt marsh provides a habitat and nursing grounds for a variety of land and marine organisms. Without the salt marsh, it is possible that some commercial fishing stocks would be harmed, or that the land ecosystems would lose much of their wildlife. The salt marsh also purifies groundwater and soaks up excess rainfall from hurricanes or other storms. Residents likely would suffer without the salt marsh, so I would not support the development. **DOK 3**

35. The icons in the outer ring are changes in Earth systems. The inner circle is the measurable changes. They are related because the outer ring causes the effects in the inner circle. **DOK 3**

36. Sample answers: Wind plays a role in moving air and weather across the planet and it also causes ocean circulation patterns. The greenhouse effect affects the amount of heat that is retained in the atmosphere and also causes melting of ice on the planet which increases ocean levels. Mountain building can block atmospheric circulation patterns such as the Andes Mountains in Chile. Underwater mountain building can also affect ocean circulation patterns by changing the topography of the ocean floor. **DOK 2**

37. Sample answer: These processes and phenomena are intertwined and there are specific interactions among the components that are not provided in the model. **DOK 2**